

Discrete geodesic interpolation

— iteratively descends along the negative **Riemannian natural-gradient** direction.

Step 1. squared-distance potential

$$\phi(q) = \frac{1}{2} \hat{d}(q, q_{\text{rand}})^2$$

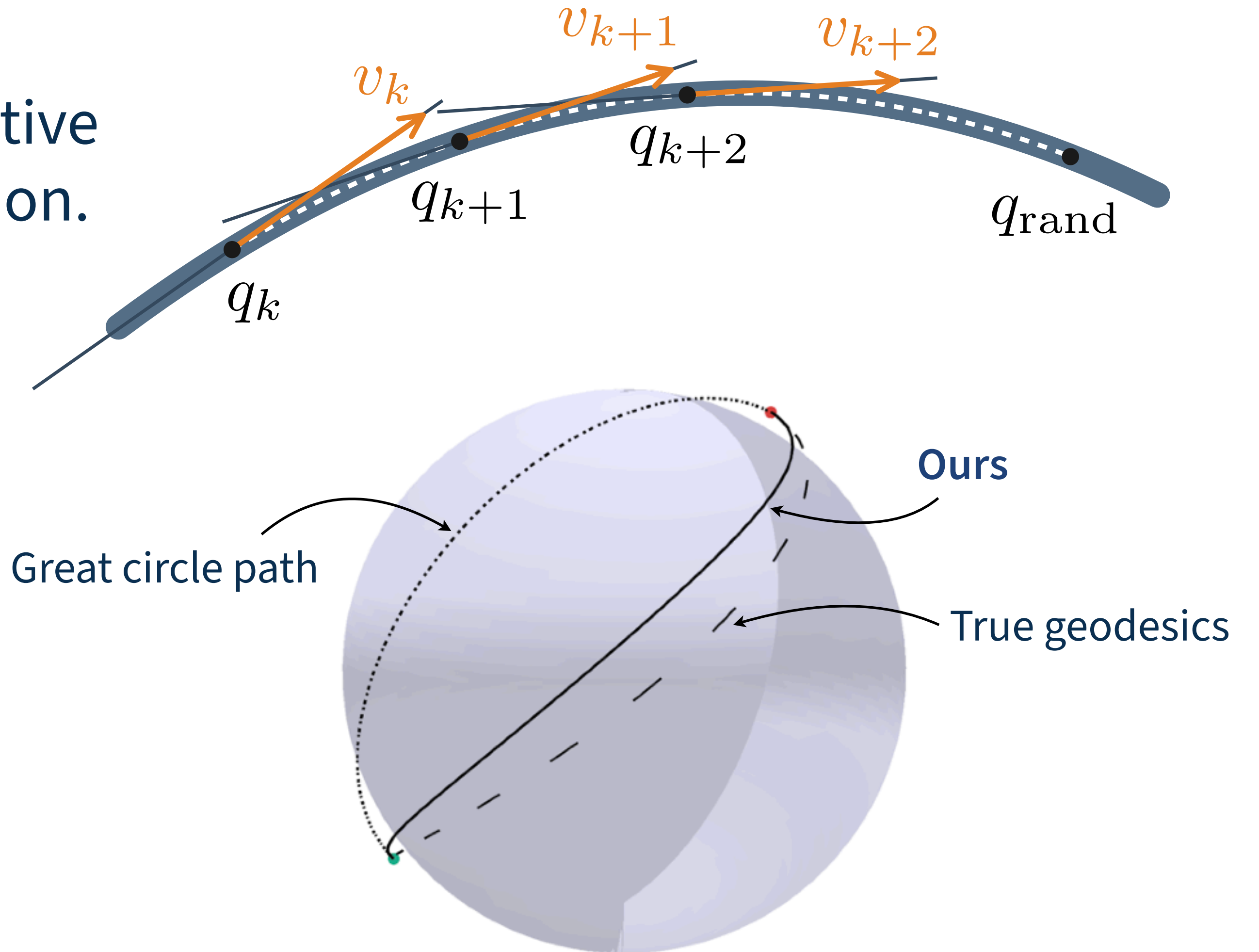
Step 2. Riemannian natural gradient

$$v_k = -G(q_k)^{-1} \nabla \phi(q_k)$$

Step 3. retraction step

$$\hat{v}_k = v_k / \|v_k\|_{q_k}$$

$$q_{k+1} = \mathcal{R}_{q_k}(s_k \hat{v}_k)$$



local planning on sphere under an anisotropic Riemannian metric

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Experiments

One framework, many systems.